

Standard Operating Procedure

Certificate Enrollment Methods & GPO Trust Distribution

Quantumspace Issuing CA — Enrollment Options, Testing, and Domain-Wide CA Trust via Group Policy

Domain: Quantumspace.com

1. Overview

This SOP covers two related but distinct topics that come after your two-tier PKI hierarchy (Root-CA → Issuing-CA → PKI-WEB-SRV) is already built and healthy in PKIView.

- Part A — how to push the Root CA and Issuing CA certificates into the trusted certificate stores of every domain-joined device using Group Policy, so clients trust certificates issued by your hierarchy.
- Part B — every practical method of actually requesting/enrolling a certificate from the Issuing CA, when to use each one, and how to test that it worked.

Note: Server names used throughout: Root-CA (offline root), Issuing-CA.Quantumspace.com (subordinate/enterprise CA), PKI-WEB-SRV.Quantumspace.com (IIS/CDP/AIA/OCSP host), AD-Domain (domain controller). Substitute your actual hostnames if different.

2. Part A – Distributing CA Certificates via GPO to All Domain-Joined Devices

2.1 Background: what already happens automatically

Because Issuing-CA is an Enterprise CA and Root-CA's certificate was published into Active Directory (certutil -dspublish RootCA, done earlier), domain-joined computers already pick up both CA certificates automatically — Windows periodically refreshes each computer's trust store from AD's Public Key Services container in the background, and this is also refreshed on every Group Policy update cycle. Most domain members will already trust your hierarchy without any extra GPO work.

Why this step: This SOP still walks through an explicit GPO-based method because: (1) it forces immediate, predictable distribution instead of waiting on the background refresh timer, (2) it gives you one place to audit/confirm exactly what's being pushed, and (3) it also works for scenarios where the automatic AD refresh is disabled or delayed by policy.

2.2 Export the CA certificates

1. On Issuing-CA, open the Certification Authority console (certsrv.msc).
2. Right-click Quantumspace Issuing CA → Properties → General tab → View Certificate → Details tab → Copy to File → export as Base-64 encoded X.509 (.CER) → save as IssuingCA.cer.
3. Repeat for the Root CA certificate: retrieve C:\Windows\System32\CertSrv\CertEnroll\Root-CA_Quantumspace Root CA.crt (already available from the earlier build) and copy it to a working folder as RootCA.cer.
4. Copy both .cer files to DC01 (or any machine with Group Policy Management installed).

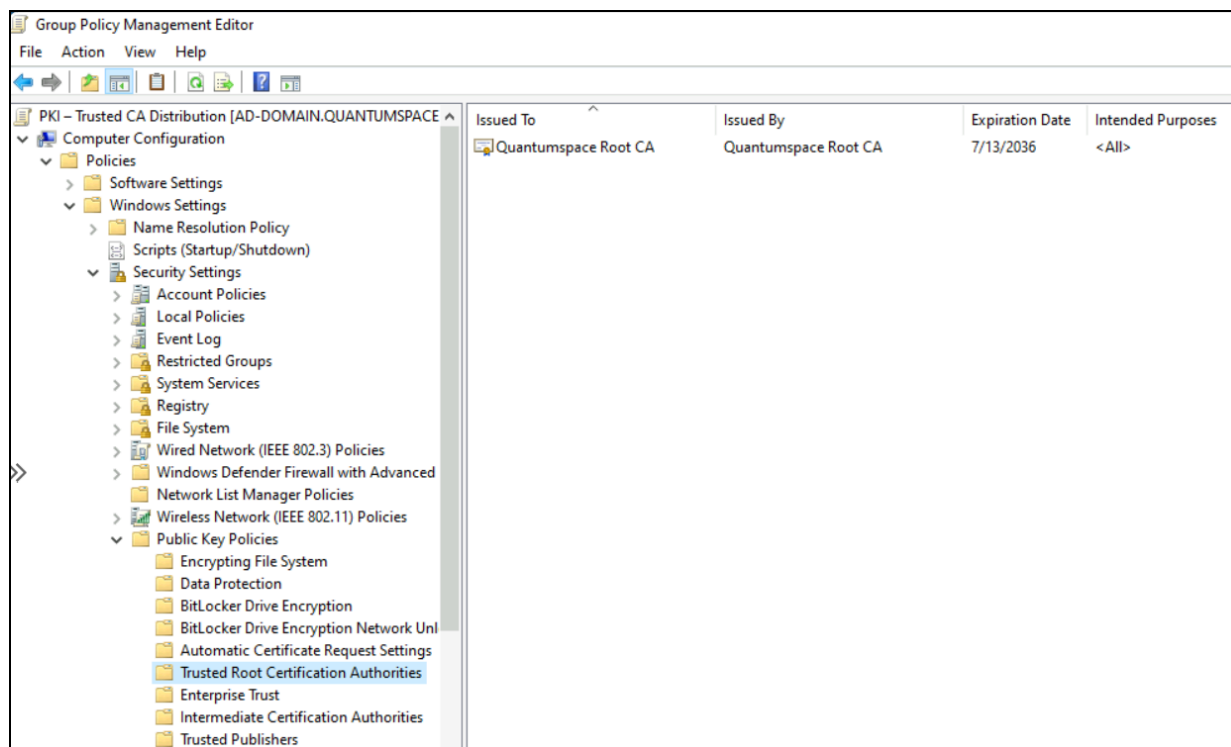
2.3 Create the GPO

1. On DC, open Group Policy Management (gpmc.msc).
2. Right-click Quantumspace.com (or the OU containing your domain computers) → Create a GPO in this domain, and Link it here.
3. Name it: PKI – Trusted CA Distribution → OK.
4. Right-click the new GPO → Edit, to open the Group Policy Management Editor.

2.4 Import the Root CA certificate into Trusted Root Certification Authorities

1. In the Group Policy Management Editor, expand Computer Configuration → Policies → Windows Settings → Security Settings → Public Key Policies.
2. Right-click Trusted Root Certification Authorities → Import.
3. Browse to RootCA.cer → Next → Next (place in Trusted Root Certification Authorities store) → Finish.

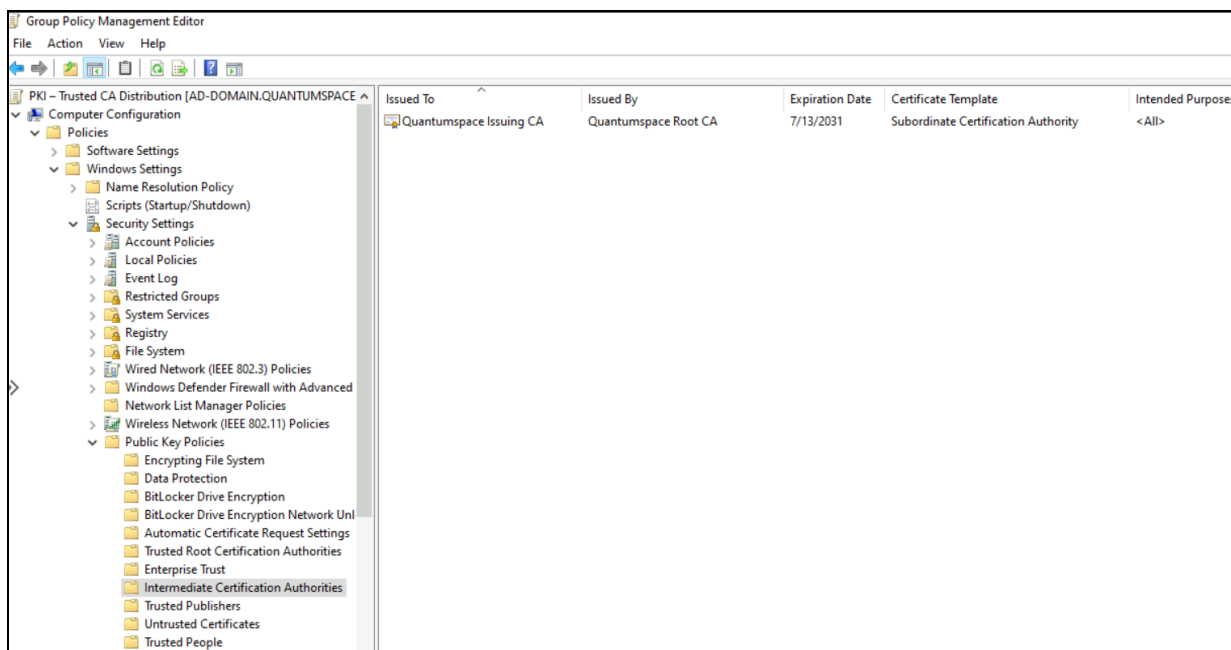
Why this step: This store is where Windows decides which CAs are inherently trustworthy as ultimate trust anchors. Every certificate chain in your PKI ultimately has to trace back to an entry here to be considered valid — this is the single most important trust decision in the whole hierarchy.



2.5 Import the Issuing CA certificate into Intermediate Certification Authorities

1. In the same editor, right-click Intermediate Certification Authorities → Import.
2. Browse to IssuingCA.cer → Next → Next (place in Intermediate Certification Authorities store) → Finish.
3. Close the Group Policy Management Editor.

Why this step: The Issuing CA's certificate deliberately does NOT go into Trusted Root — it is not a root, it is a subordinate. Placing it in Intermediate Certification Authorities lets Windows build the full chain (leaf certificate → Issuing CA → Root CA) without treating the Issuing CA itself as a standalone trust anchor. Mixing this up (e.g., accidentally importing the Issuing CA cert into Trusted Root) is a common misconfiguration that weakens the trust model.



2.6 Link, scope, and force the update

1. Confirm the GPO is linked at the domain root (or the OU covering all the machines you want to trust the hierarchy).
2. On a test client, run:

gpupdate /force

1. Restart the client if prompted, though computer-configuration security settings like this usually apply without a reboot.

2.7 Testing / Verification

Test / Verify: On the test client, run mmc → Add Certificates snap-in (Computer account) → expand Trusted Root Certification Authorities → Certificates, and confirm Quantumspace Root CA is listed. Then expand Intermediate Certification Authorities → Certificates and confirm Quantumspace Issuing CA is listed.

The first screenshot shows a command prompt window with the following commands and output:

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.5256]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>hostname
SCCM-Primary

C:\Windows\system32>nslookup SCCM-Primary
Server: ad-domain.internal.cloudapp.net
Address: 10.2.0.4

Name: SCCM-Primary.Quantumspace.com
Address: 10.3.1.4

C:\Windows\system32>gpupdate
Updating policy...
Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Windows\system32>mmc
C:\Windows\system32>
```

The second screenshot shows the MMC console for Trusted Root Certification Authorities. The table below represents the data visible in the console:

Issued To	Issued By	Expiration Date	Intended Purposes
DigiCert Global Root G2	DigiCert Global Root G2	1/15/2038	<All>
DigiCert Global Root G3	DigiCert Global Root G3	1/15/2038	<All>
Microsoft Authenticode(tm) Root...	Microsoft Authenticode(tm) Root...	12/31/1999	Secure Email, Code...
Microsoft ECC Product Root Ce...	Microsoft ECC Product Root Certi...	2/27/2043	<All>
Microsoft ECC Root Certificate ...	Microsoft ECC Root Certificate Au...	7/18/2042	<All>
Microsoft ECC TS Root Certifica...	Microsoft ECC TS Root Certificate ...	2/27/2043	<All>
Microsoft Root Authority	Microsoft Root Authority	12/31/2020	<All>
Microsoft Root Certificate Auth...	Microsoft Root Certificate Authori...	5/9/2021	<All>
Microsoft Root Certificate Auth...	Microsoft Root Certificate Authori...	6/23/2035	<All>
Microsoft Root Certificate Auth...	Microsoft Root Certificate Authori...	3/22/2036	<All>
Microsoft RSA Root Certificate ...	Microsoft RSA Root Certificate Au...	7/18/2042	<All>
Microsoft Time Stamp Root Cer...	Microsoft Time Stamp Root Certif...	10/22/2039	<All>
NO LIABILITY ACCEPTED, (c)197...	NO LIABILITY ACCEPTED, (c)197 Ve...	1/7/2004	Time Stamping
Quantumspace Root CA	Quantumspace Root CA	7/13/2036	<All>
Quantumspace Root CA	Quantumspace Root CA	7/13/2036	<All>
Symantec Enterprise Mobile Ro...	Symantec Enterprise Mobile Root ...	3/14/2032	Code Signing
Thawte Timestamping CA	Thawte Timestamping CA	12/31/2020	Time Stamping

The third screenshot shows a command prompt window with the following commands and output:

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.5256]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>hostname
SCCM-Primary

C:\Windows\system32>nslookup SCCM-Primary
Server: ad-domain.internal.cloudapp.net
Address: 10.2.0.4

Name: SCCM-Primary.Quantumspace.com
Address: 10.3.1.4

C:\Windows\system32>gpupdate
Updating policy...
Computer Policy update has completed successfully.
User Policy update has completed successfully.

C:\Windows\system32>mmc
C:\Windows\system32>
```

The fourth screenshot shows the MMC console for Intermediate Certification Authorities. The table below represents the data visible in the console:

Issued To	Issued By	Expiration Date	Intended Purposes
Microsoft Windows Hardware ...	Microsoft Root Authority	12/31/2002	Code Signing, Win...
Quantumspace Issuing CA	Quantumspace Root CA	7/13/2031	<All>
Quantumspace Issuing CA	Quantumspace Root CA	7/13/2031	<All>
Quantumspace Root CA	Quantumspace Root CA	7/13/2036	<All>
Root Agency	Root Agency	12/31/2039	<All>
www.verisign.com/CPS Incorp...	Class 3 Public Primary Certificatio...	10/24/2016	Server Authenticati...

Test / Verify: Command-line alternative, run on the client: certutil -store root (look for Quantumspace Root CA) and certutil -store CA (look for Quantumspace Issuing CA).

Test / Verify: Full chain test: take any certificate previously issued by Issuing-CA and run certutil -verify -urlfetch <file>.cer on the client — the output should show a clean chain to Quantumspace Root CA with no untrusted-root errors.

Note: If the certificates don't appear after gpupdate /force, run gpresult /h report.html on the client and confirm the 'PKI – Trusted CA Distribution' GPO shows as Applied, not Denied/Filtered — a common cause is the GPO being linked to the

wrong OU, or Security Filtering excluding the test machine's computer account.

FileC:/Windows/System32/report.html

Registry

Success

297 Millisecond(s)

7/13/2026 11:16:53 PM

View Log

Security

Success

922 Millisecond(s)

7/2/2026 2:56:01 PM

View Log

Settings

Policies

Windows Settings

Security Settings

Account Policies/Password Policy

Account Policies/Account Lockout Policy

Local Policies/Security Options

Public Key Policies/Certificate Services Client - Auto-Enrollment Settings

Public Key Policies/Encrypting File System

Public Key Policies/Trusted Root Certification Authorities

Certificates

Issued To	Issued By	Expiration Date	Intended Purposes	Winning GPO
Quantumspace Root CA	Quantumspace Root CA	7/13/2036 8:11:43 PM	<All>	PKI - Trusted CA Distribution

For additional information about individual settings, launch the Local Group Policy Object Editor.

Public Key Policies/Intermediate Certification Authority Certificates

Issued To	Issued By	Expiration Date	Intended Purposes	Winning GPO
Quantumspace Issuing CA	Quantumspace Root CA	7/13/2031 8:50:58 PM	<All>	PKI - Trusted CA Distribution

For additional information about individual settings, launch the Local Group Policy Object Editor.

Activate Windows

3. Part B – Certificate Enrollment Methods from the Issuing CA

The table below summarizes every practical way to get a certificate from Quantumspace Issuing CA. Detailed steps and a test for each follow.

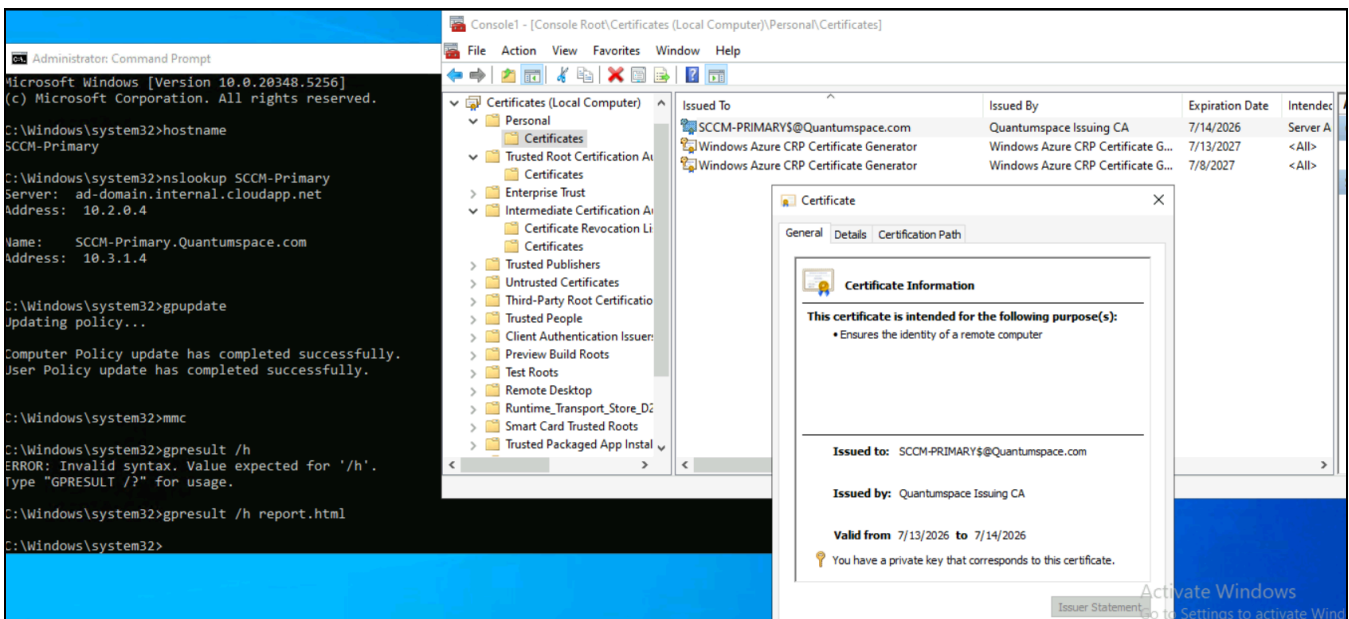
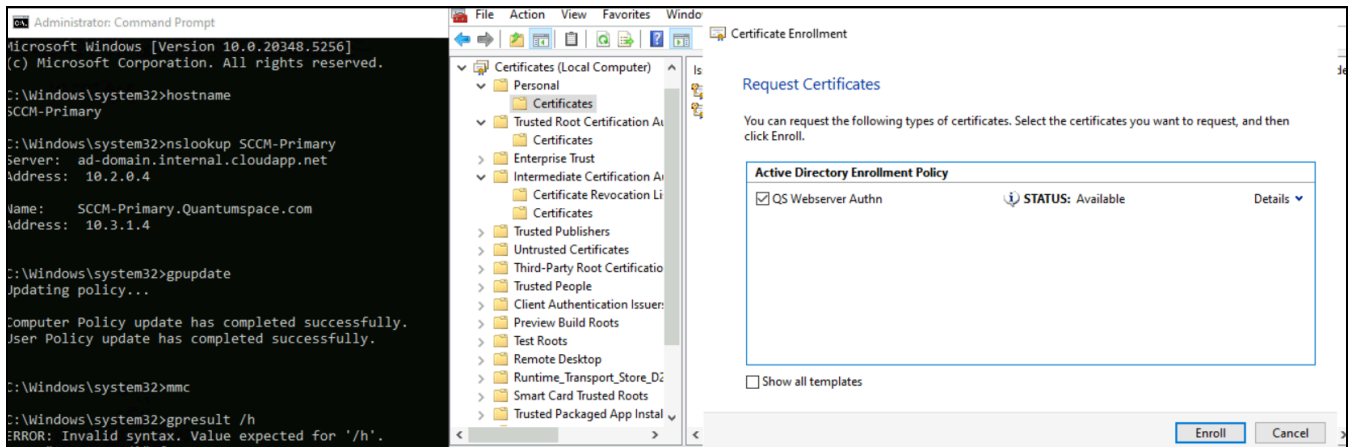
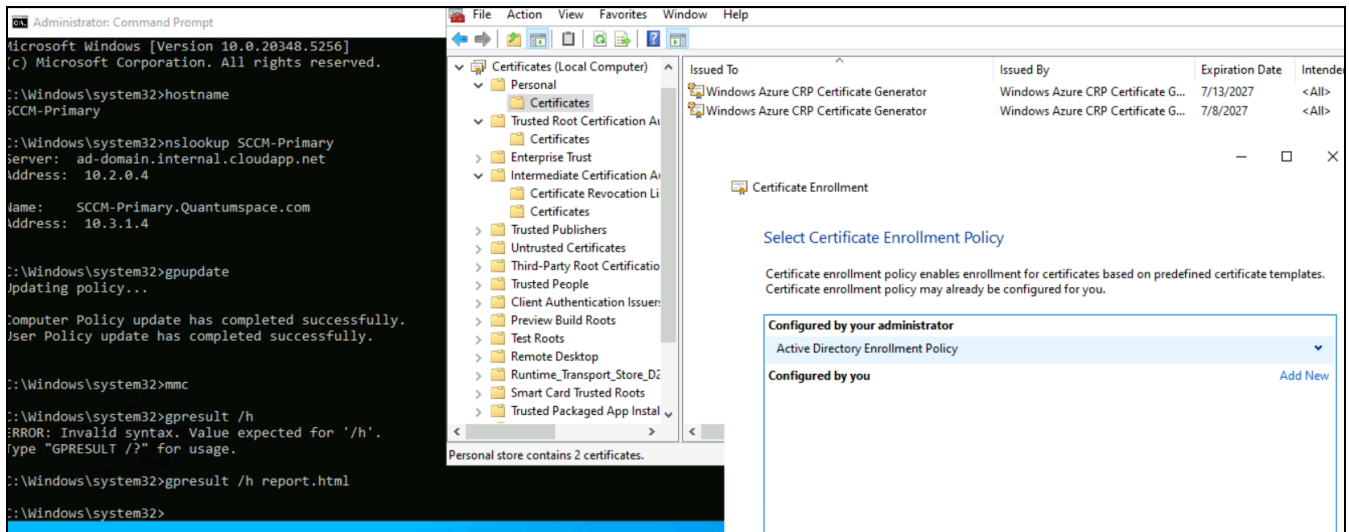
Method	Best for	Requires domain join?	Admin effort
MMC Certificates Snap-in	One-off manual requests, troubleshooting, admin testing	Yes (Enterprise templates)	Low, manual per machine
Web Enrollment (certsrv)	Non-domain devices, manual PKCS#10/CSR-based requests, admins without RSAT	No (but slower/manual approval possible)	Low, manual per request
Autoenrollment (GPO)	Fleet-wide, zero-touch issuance/renewal for users & computers	Yes	Set up once, then hands-off
certreq (CLI / INF)	Scripted/automated requests, servers, non-interactive scenarios	Optional	Medium, scriptable
PowerShell (Get-Certificate / PKI module)	Automation, DSC, CI/CD pipelines, bulk operations	Optional	Medium, scriptable
NDES / SCEP	Non-domain, mobile, network devices (routers, printers, MDM-enrolled phones)	No	High, extra role + policy design

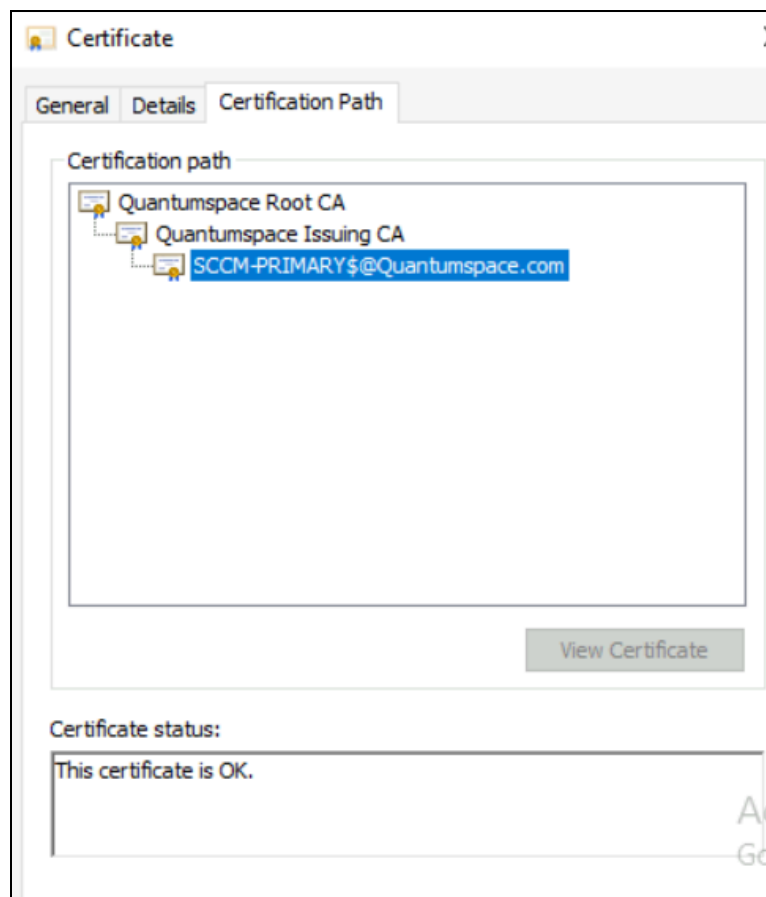
3.1 Method 1 – MMC Certificates Snap-in (interactive, manual)

Why this step: This is the standard way an administrator manually requests a certificate for a specific domain-joined computer or user, on demand, without waiting for autoenrollment. Good for testing a new template or issuing a one-off certificate.

1. On the target domain-joined machine, run mmc → File → Add/Remove Snap-in → Certificates → choose Computer account (or My user account for a user cert) → Finish → OK.

- Expand Certificates → Personal → right-click → All Tasks → Request New Certificate.
- Click Next past the Before You Begin and Select Certificate Enrollment Policy screens (Active Directory Enrollment Policy should be pre-selected).
- Select the desired template (e.g., Computer or Workstation Authentication) from the list → if required fields are marked, click More information is required ... and fill in Subject Name details → OK.
- Click Enroll → Finish.





Test / Verify: In the same MMC, refresh the Personal → Certificates node and confirm the new certificate appears, with Issued By showing Quantumspace Issuing CA. Double-click it → Certification Path tab → confirm the path shows Quantumspace Root CA → Quantumspace Issuing CA → the new certificate, all with a green checkmark and no red/yellow warning icon.

certsrv - [Certification Authority (Local)\Quantumspace Issuing CA\Issued Certificates]

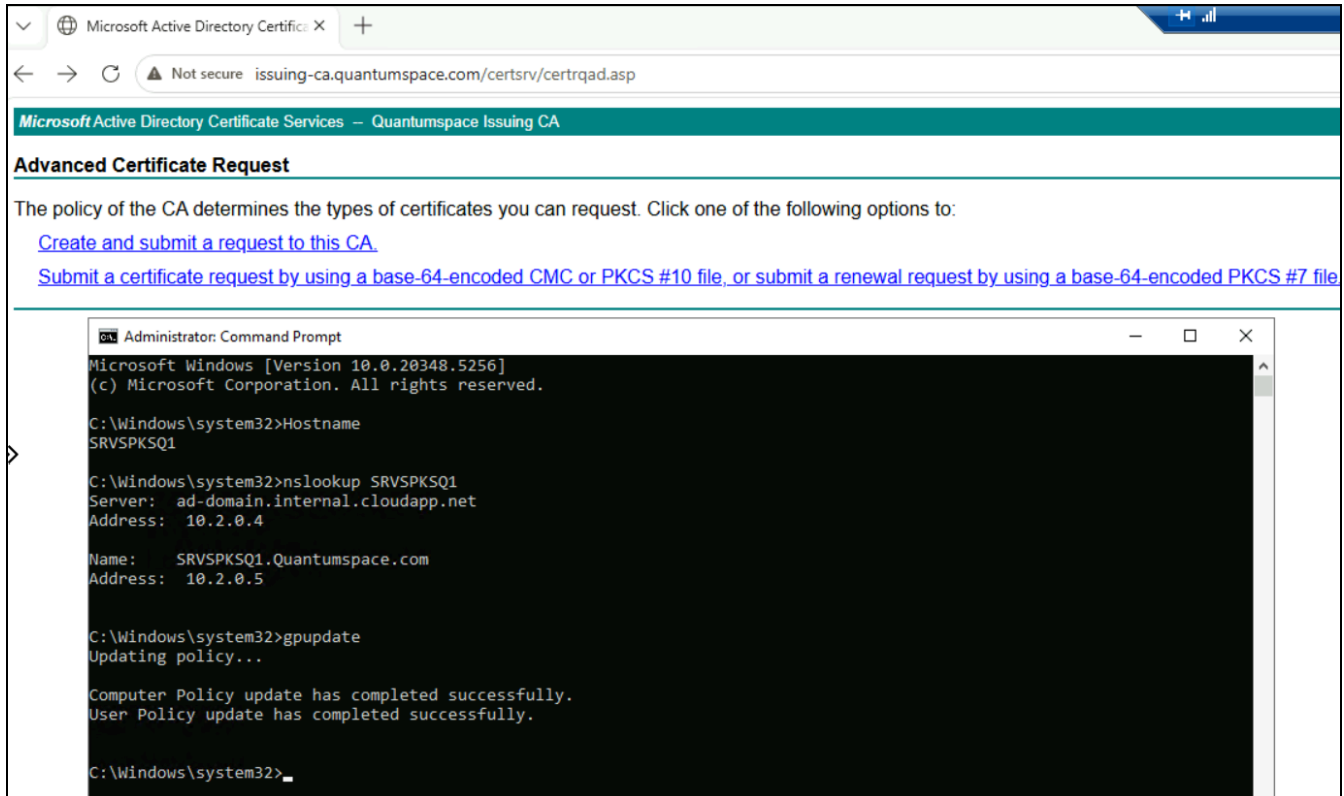
Request ID	Requester Name	Binary Certificate	Certificate Template	Serial Number	Certificate Effective Date	Certificate Expiration Date
3	QUANTUMSPACE\ISSUING-CAS	-----BEGIN CERTI...	CA Exchange (CAExchange)	1d0000000337a...	7/13/2026 9:02 PM	7/20/2026 9:12 PM
4	QUANTUMSPACE\SCCM-PRIMARY\$	-----BEGIN CERTI...	QS Webserver Authn (1.3.6.1.4.1.311.21.8.9273...	1d00000004aec...	7/13/2026 11:35 PM	7/14/2026 11:35 PM

3.2 Method 2 – Certificate Services Web Enrollment

Why this step: This is the only method that works for non-domain-joined machines, or for admins/users who don't have RSAT/MMC access. It relies on the Certification Authority Web Enrollment role service you already installed on Issuing-CA.

1. From any machine that can reach the domain network, open a browser to: <http://Issuing-CA.Quantumspace.com/certsrv>
2. Log in with domain credentials if prompted.
3. Click Request a certificate → advanced certificate request.

4. Either paste a PKCS#10 CSR generated elsewhere, or (on Internet Explorer/legacy-compatible browsers only) use the basic form; modern browsers generally require the CSR-paste method since client-side key generation via ActiveX is deprecated.
5. Select the Certificate Template from the dropdown → Submit.
6. If the template requires manual approval, the request will show Certificate Pending — an administrator must approve it in the Certification Authority console (Pending Requests → right-click → Issue) before it can be downloaded.
7. Return to <http://Issuing-CA.Quantumspace.com/certsrv/certmgr.asp>, view the request status, and download the certificate/chain.



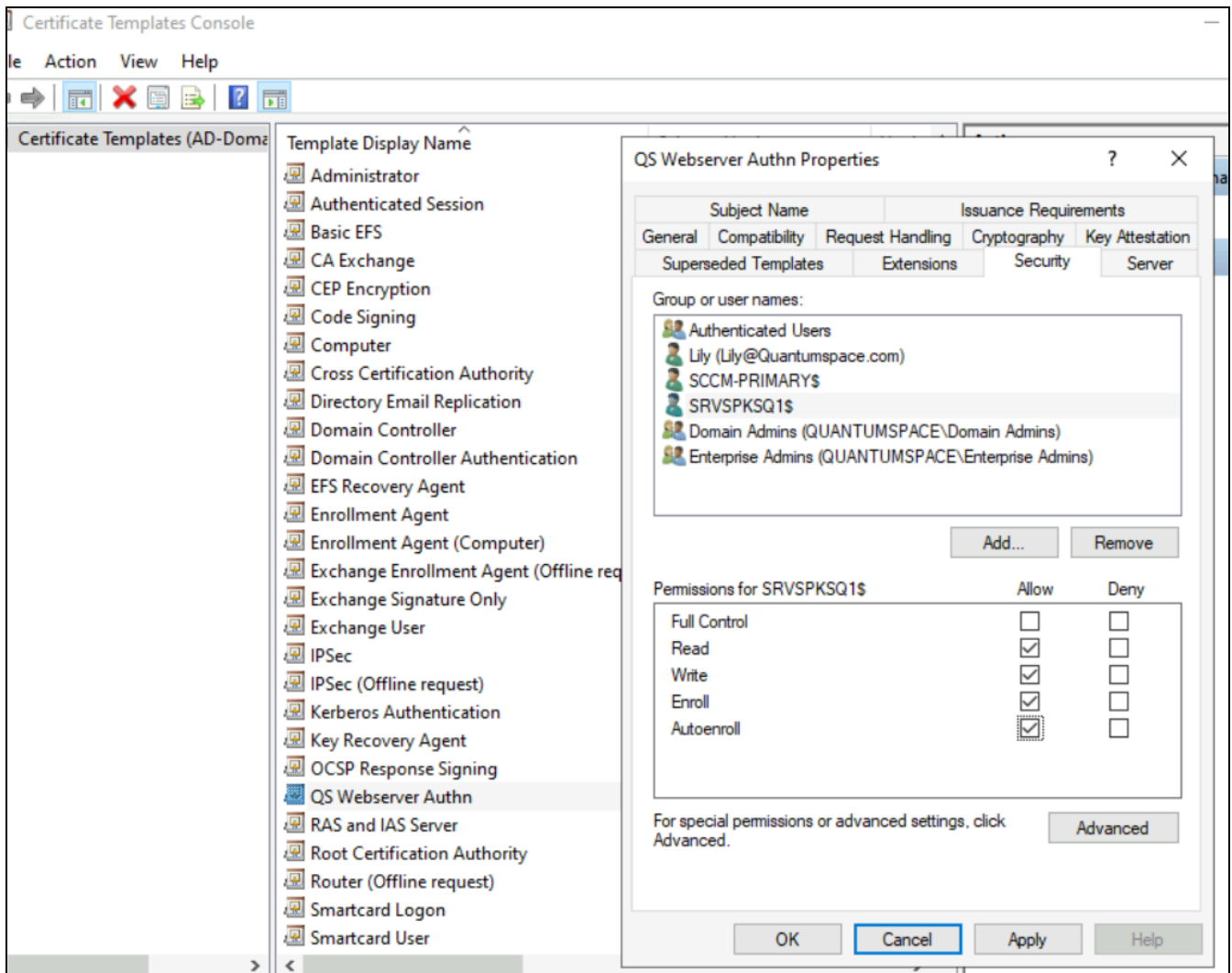
Test / Verify: Open the downloaded .cer file and confirm the Issued By and Issued To fields are correct, and that Windows shows 'This certificate is OK' when the destination machine already trusts the Root/Issuing CA (per Part A). If it shows untrusted, that machine hasn't received the GPO trust distribution yet.

3.3 Method 3 – Autoenrollment via Group Policy (recommended for fleet-wide issuance)

Why this step: This is the method you want running permanently for ordinary domain computers and users — certificates are requested, issued, installed, and renewed automatically with zero end-user or admin interaction per device.

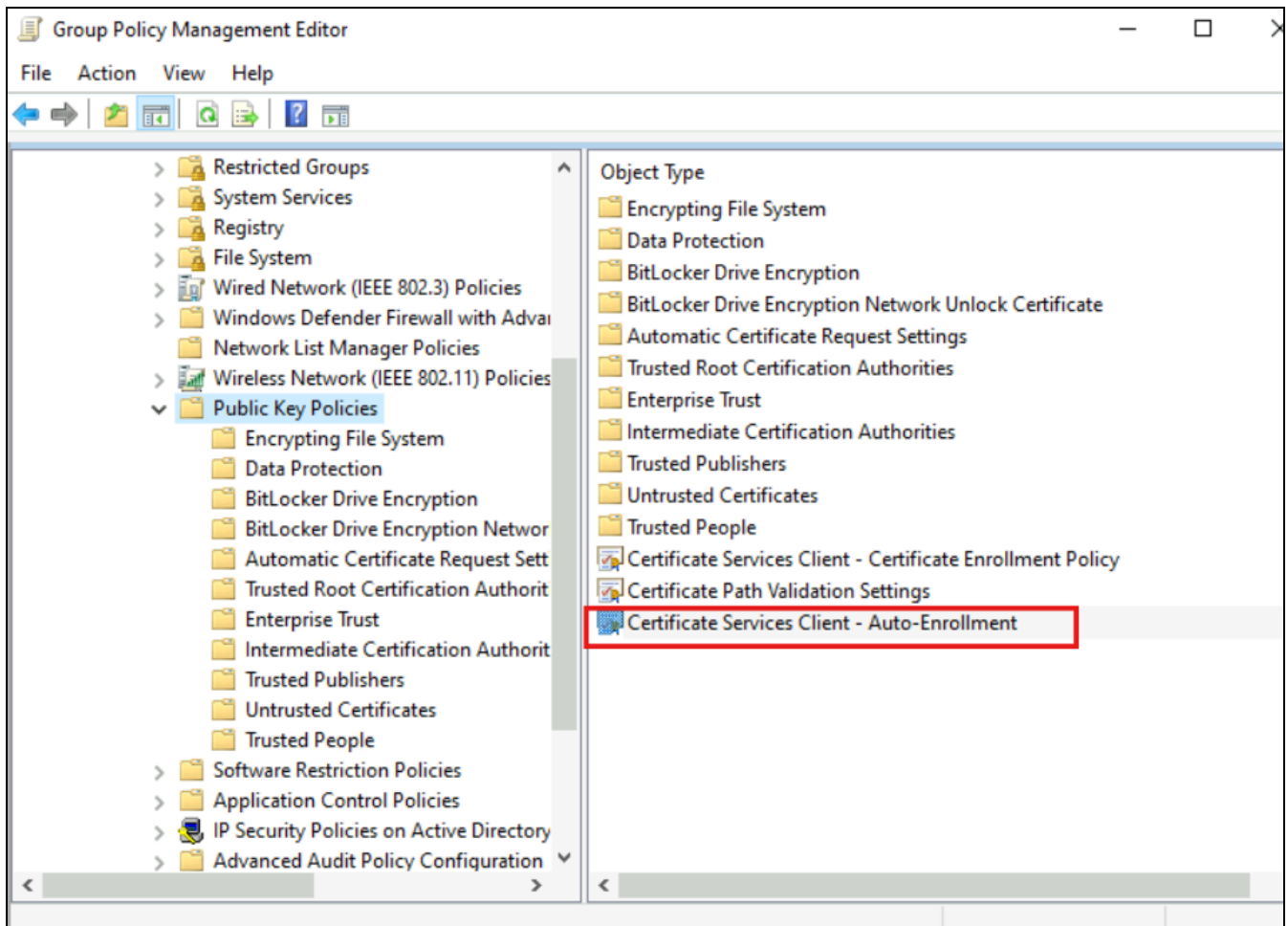
3.3.1 Configure the certificate template for autoenrollment

1. On Issuing-CA, open `certsrv.msc` → Certificate Templates → Manage (opens the Certificate Templates console).
2. Right-click the template you want to autoenroll (e.g., Computer or a duplicated custom template) → Properties → Security tab.
3. Select the relevant group (e.g., Domain Computers) → check Allow for both Enroll and Autoenroll → OK.
4. Back in `certsrv.msc`, right-click Certificate Templates → New → Certificate Template to Issue → select the template (skip if already issued) → OK.

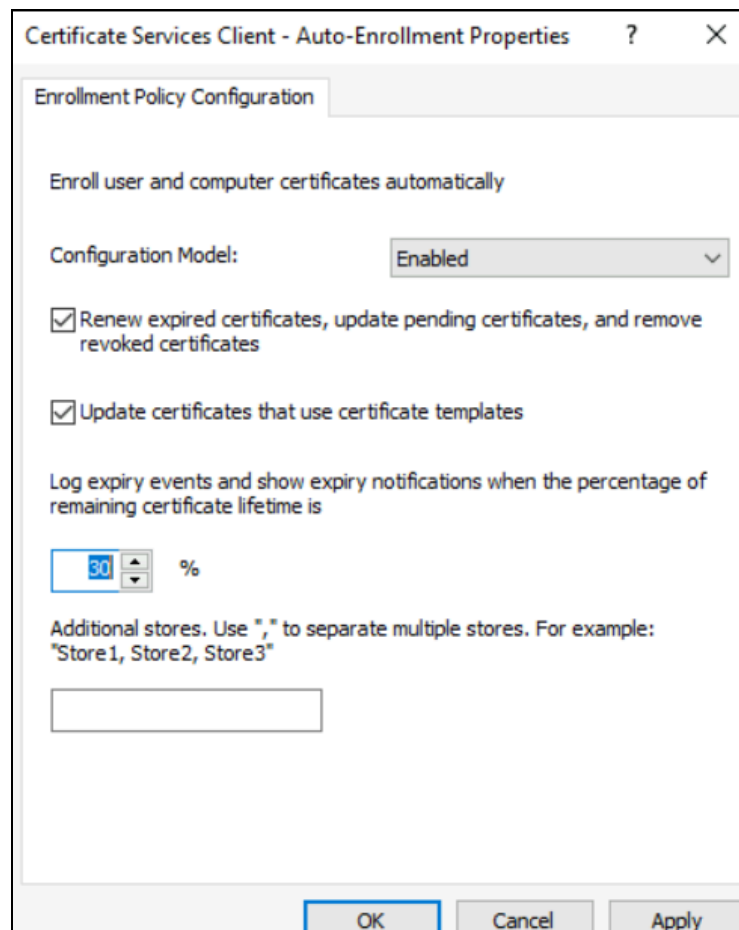


3.3.2 Enable autoenrollment via GPO

1. On DC, in Group Policy Management, edit an existing GPO (or create PKI – Autoenrollment) linked at the domain/OU level.
2. Computer Configuration → Policies → Windows Settings → Security Settings → Public Key Policies → double-click Certificate Services Client - Auto-Enrollment.



1. Set Configuration Model to Enabled, check Renew expired certificates, update pending certificates, and remove revoked certificates and Update certificates that use certificate templates → OK.



1. For user certificate templates, repeat the same setting under User Configuration → Policies → Windows Settings → Security Settings → Public Key Policies.
2. On a target client, run:

```
gpupdate /force
```

```
certutil -pulse
```

Test / Verify: On the client, open mmc → Certificates (Computer account) → Personal → Certificates and confirm the certificate now appears without having requested it manually. Check Issued By = Quantumspace Issuing CA and that the Valid from date matches roughly when gpupdate was run.

Windows Settings

Security Settings

Group Policy Objects

Applied GPOs

Default Domain Policy [{31B2F340-016D-11D2-945F-00C04FB984F9}]

PKI – Autoenrollment [{2200F2E3-DFFF-42D1-B2C3-C28474191C82}]

Link Location

Extensions Configured

Enforced

Disabled

Security Filters

Revision

WMI Filter

Quantumspace.com

{B1BE8D72-6EAC-11D2-A4EA-00C04F79F83A}

Registry

Yes

None

NT AUTHORITY\Authenticated Users

AD (8), SYSVOL (8)

Before:

Console1 - [Console Root\Certificates (Local Computer)\Personal\Certificates]

File Action View Favorites Window Help

Console Root

Certificates (Local Computer)

Personal

Certificates

Trusted Root Certification Authorities

Enterprise Trust

Intermediate Certification Authorities

Issued To

Issued By

Expiration Date

Intended Purposes

SCCM-PRIMARYS@Quantumspace.com

Quantumspace Issuing CA

7/14/2026

Server Authentication

Windows Azure CRP Certificate Generator

Windows Azure CRP Certificate G...

7/15/2027

<All>

Windows Azure CRP Certificate Generator

Windows Azure CRP Certificate G...

7/13/2027

<All>

After:

Certification Authority (Local)

Quantumspace Issuing CA

Revoked Certificates

Issued Certificates

Pending Requests

Failed Requests

Certificate Templates

Request ID

Requester Name

Binary Certificate

Certificate Template

Serial Number

Certificate Effective Date

Certificate Expiration Date

4

QUANTUMSPACE\ISSUING-CAS

-----BEGIN CERTI...

CA Exchange (CAEx...

1d0000000337aa540a0558097300000000...

7/13/2026 9:02 PM

7/20/2026 9:12 PM

3

QUANTUMSPACE\SCCM-PRIMARYS

-----BEGIN CERTI...

QS Webserver Authn...

1d00000004aec56eb4c0ea48f1000000000...

7/13/2026 11:35 PM

7/14/2026 11:35 PM

5

QUANTUMSPACE\SRVSPKQS1\$

-----BEGIN CERTI...

QS Webserver Authn...

1d00000005a4d3e7d57ecc512000000000...

7/14/2026 12:40 AM

7/15/2026 12:40 AM

6

QUANTUMSPACE\SRVSPKQS1\$

-----BEGIN CERTI...

QS Webserver Authn...

1d00000006812596e8639f5b16000000000...

7/15/2026 8:00 PM

7/16/2026 8:00 PM

7

QUANTUMSPACE\SCCM-PRIMARYS

-----BEGIN CERTI...

QS Webserver Authn...

1d00000007f0ac3afebcb8a9e0000000000...

7/15/2026 7:59 PM

7/16/2026 7:59 PM

File Action View Favorites Window Help

Console Root

Certificates (Local Computer)

Personal

Certificates

Trusted Root Certification Authorities

Enterprise Trust

Intermediate Certification Authorities

Issued To

Issued By

Expiration Date

Certificate Template

Intended Purposes

SCCM-Primary.Quantumspace.com

Quantumspace Issuing CA

7/16/2026

QS Webserver Authn

Server Authenticati...

Windows Azure CRP Certificate Generator

Windows Azure CRP Certificate Gener...

7/15/2027

<All>

<All>

Windows Azure CRP Certificate Generator

Windows Azure CRP Certificate Gener...

7/13/2027

<All>

<All>

Test / Verify: Confirm renewal behavior works by checking Event Viewer → Applications and Services Logs → Microsoft → Windows → CertificateServicesClient-Lifecycle-System for autoenrollment success events (Event ID 1006/1007

typically indicate successful enrollment).

- Configure Server Certificate Autoenrollment

3.4 Method 4 – certreq (command-line / INF-based request)

Useful for scripted deployments, servers with no GUI, or requesting a certificate with very specific Subject/SAN values not easily entered through the MMC wizard.

1. On the target machine, create a request INF file, e.g. C:\Temp\request.inf, with content such as:

[Version]

Signature="\$Windows NT\$"

[NewRequest]

Subject = "CN=webapp01.quantumspace.com"

KeyLength = 3072

KeySpec = 1

Exportable = FALSE

MachineKeySet = TRUE

SMIME = FALSE

PrivateKeyArchive = FALSE

UserProtected = FALSE

UseExistingKeySet = FALSE

ProviderName = "Microsoft RSA SChannel Cryptographic Provider"

ProviderType = 12

RequestType = PKCS10

KeyUsage = 0xa0

[EnhancedKeyUsageExtension]

OID=1.3.6.1.5.5.7.3.1 ; Server Authentication

[RequestAttributes]

CertificateTemplate = WebServer

1. Generate the request:

```
certreq -new C:\Temp\request.inf C:\Temp\request.req
```

1. Submit it to the Issuing CA:

```
certreq -submit -config "Issuing-CA.Quantumspace.com\Quantumspace Issuing CA" C:\Temp\request.req  
C:\Temp\certnew.cer
```

1. If the template requires manual approval, retrieve it later once issued (use the RequestID shown at submission):

```
certreq -retrieve <RequestID> C:\Temp\certnew.cer
```

1. Install the issued certificate and bind it to the private key that was generated with the request:

```
certreq -accept C:\Temp\certnew.cer
```

Test / Verify: Run `certutil -store My` on the machine and confirm the new certificate appears bound to a private key (look for a line indicating the private key is present, not just the public certificate).

3.5 Method 5 – PowerShell (Get-Certificate)

Useful for automation, Desired State Configuration, or bulk provisioning where you want enrollment as part of a repeatable script rather than a one-off manual action.

1. On the target machine (must have line of sight to Issuing-CA and, ideally, be domain-joined for template-based auto-discovery), open an elevated PowerShell session.
2. Request a certificate directly against a template:

```
Get-Certificate -Template "WebServer" -CertStoreLocation "Cert:\LocalMachine\My" -SubjectName  
"CN=webapp02.quantumspace.com" -DnsName "webapp02.quantumspace.com"
```

1. If the request is pending manual approval, the cmdlet returns a status of Pending; re-run it later, or check status directly:

```
Get-Certificate -Template "WebServer" -CertStoreLocation "Cert:\LocalMachine\My"
```

Test / Verify: Run `Get-ChildItem Cert:\LocalMachine\My` in PowerShell and confirm the new certificate is listed, with `.Issuer` showing `CN=Quantumspace Issuing CA` and `.HasPrivateKey` returning `True`.

- [Get-Certificate cmdlet reference](#)

3.6 Method 6 (Optional/Advanced) – NDES / SCEP for Non-Domain and Mobile Devices

None of the previous methods work for devices that can't run Windows enrollment clients or join the domain — network printers, routers, firewalls, or MDM-managed mobile phones. NDES bridges this gap by exposing SCEP, a protocol those devices already understand.

- Install the Network Device Enrollment Service (NDES) role service, typically on a dedicated member server (not directly on Issuing-CA, for isolation).
- NDES requires a dedicated service account and issues certificates using a fixed template mapped internally to IPsec (Offline request) — plan template mapping carefully as SCEP has no interactive template-selection step.
- Once configured, devices/MDM platforms (e.g., Intune) are pointed at the NDES SCEP URL to request certificates automatically on enrollment.

Note: This method involves materially more setup (dedicated server, service account, IIS configuration, and security hardening since NDES is often internet-facing for mobile scenarios) than the other five. Treat it as a separate follow-up project rather than a quick lab add-on; it's included here for completeness.

- [Network Device Enrollment Service \(NDES\) role overview](#)

4. Common Validation Checklist (any method)

- Certification Path shows all three levels (leaf → Quantumspace Issuing CA → Quantumspace Root CA) with green checkmarks.
- `certutil -verify -urlfetch <cert file>` completes with no revocation or chain errors.
- The certificate's Enhanced Key Usage matches its intended purpose (e.g., Server Authentication for a web server cert, Client Authentication for a user/device cert).
- On a machine that has NOT yet received the Part A GPO, the same certificate should show as untrusted — confirming that trust is coming from your GPO/AD publishing rather than a manually-added exception.
- Revocation checking succeeds (no 'revocation status unknown' warnings) — confirms CDP/AIA/OCSP from the earlier SOP are still functioning correctly for newly issued certificates.

5. Reference Links

- [Configure Certificate Autoenrollment](#)
- [Get-Certificate \(PowerShell PKI module\)](#)
- [Certutil](#)
- [gpupdate](#)